# f00tils

*Freestanding assembly GNU userland * https://f00.sh*

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**STANDARD OPERATING PROCEDURE**

| | |
|---|---|
| **Document:** | SOP-F00-OPS-001 |
| **Date:** | 2026-07-27 |
| **Version:** | v0.16.8 |
| **Standard:** | NASA NAII 1450.3 memorandum layout |

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**Reply to Attn of:** OPS/F00

| | |
|---|---|
| **TO:** | Maintainers and Release Operators |
| **FROM:** | Project Engineering |
| **SUBJECT:** | f00tils operator standard operating procedure (clone, build, gates, install, release) |
| **REF:** | AGENTS.md, README.md, install.sh, docs/MODERN-FEATURES.md, v0.16.8 |

Bottom line: --core is never auto-detected from scripts or non-TTY. Pipes only suppress chrome. For GNU-identical PATH drop-in use export F00_CORE=1, config core=true, or pass --core. Build with cd asm && make && make check. Regenerate screenshots every release with scripts/render-brand-assets.py.

### 1. Purpose

This memorandum is the living operator handbook for f00tils (binary f00 / tools f00-*). It states how to clone the repository, build the freestanding multicall, run quality gates, install, and ship a release.

### 2. Scope

This SOP covers full project operations. It does not replace man pages, README narrative, or the scoreboard on https://f00.sh. Those remain the public ship surfaces.

- repository layout and product laws
- host prerequisites
- x86-64 build and boring-solid gates
- hot-path and speed law
- aarch64 freestanding port
- local and end-user install
- release, packaging, and site install sync

- user-facing documentation rules

## 3. Product identity

| Item | Value |
| --- | --- |
| Product name | f00tils |
| Binary / multicall | f00 |
| Tool names | f00-* (bare names when replace is on) |
| Site | https://f00.sh |
| Repository | github.com/theesfeld/f00 |
| License | MIT only |
| Primary target | Linux x86-64 freestanding (NASM, no libc) |
| Port path | Linux aarch64 freestanding (GNU as, asm/port/aarch64) |
| Surface | 115 tools (coreutils + grep + findutils + diffutils) |

## 4. Product laws (non-negotiable)

Language purity: no Rust, C application code, libc, or polyglot product dependencies in first-party product code under asm/. Shell is allowed only for bootstrap, install, packaging, and benches.

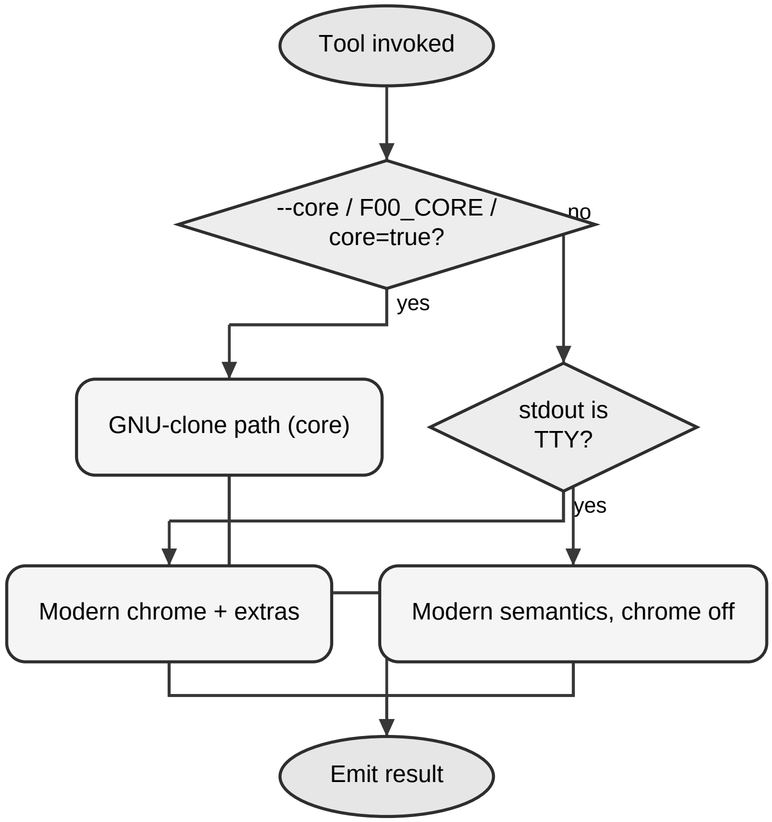
4.1 Clone first (--core). Every covered GNU tool has a f00-* counterpart. Under --core, match the GNU tool for scripts: flags, exit codes, and byte-identical stdout/stderr for the same inputs.

4.2 --core must win on resources. On the core path, f00 must beat the GNU tool on wall time and CPU (user+sys). Correct-but-slower is not done.

4.3 Modern is amazing (default). Non--core is never a pale GNU subset: themed chrome, better layout, icons where relevant, --json/--csv, and extra power.

4.4 One binary. Multicall by argv0 (f00-grep, grep, ...).

*Figure. Script vs interactive path*




## 5. Repository layout

| Path | Role |
| --- | --- |
| asm/ | Product source, Makefile, man pages, benches |
| asm/port/aarch64/ | aarch64 freestanding multicall |
| site/ | https://f00.sh (GitHub Pages) and site install.sh |
| install.sh | Root installer (must match site/ and scripts/) |
| packaging/ | AUR, nfpm (deb/rpm/arch) |
| Formula/ | Homebrew formula |
| docs/ | Compliance, UX, progress, this SOP |
| file_id.diz | Release scene card (GitHub Release asset only) |
| scripts/ | Release, package, and bench generators |

## 6. Prerequisites

6.1 x86-64 host (primary): Linux x86-64; nasm; ld (binutils); make; python3; GNU coreutils/grep/findutils/diffutils on PATH for parity and speed comparison.

6.2 aarch64 port (optional): aarch64-linux-gnu-as and aarch64-linux-gnu-ld; qemu-aarch64-static (or equivalent) for smoke on x86-64 hosts.

## 7. Procedure - clone and build (x86-64)

Do this on every new worktree before you edit product code. Root convenience targets: make (asm), make check, make aarch64, make install.

7.1 Clone or update the repository.

```
git clone git@github.com:theesfeld/f00.git
cd f00
git checkout main
git pull --ff-only
```

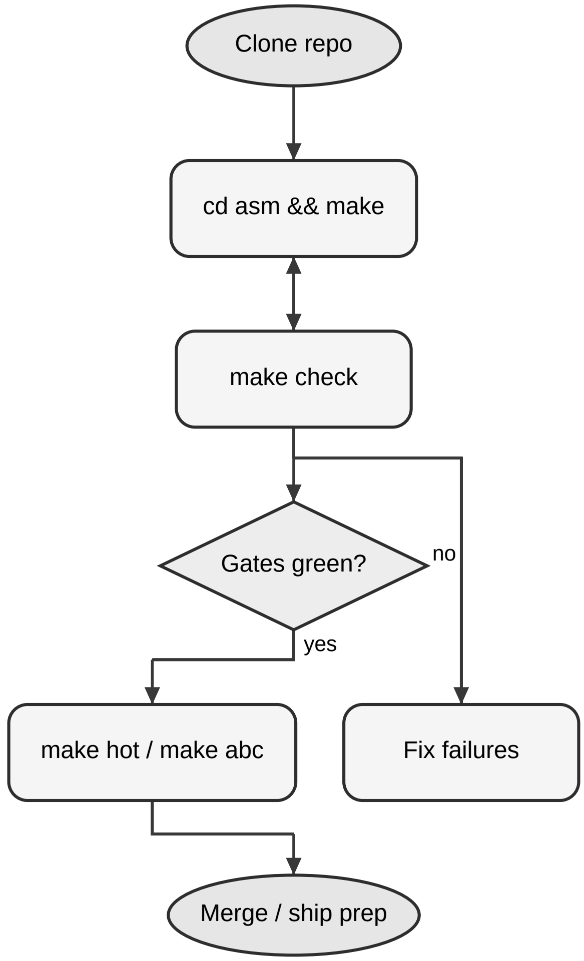
7.2 Enter the product tree and build the multicall.

```
cd asm
make
```

7.3 Confirm the binary runs.

```
./f00 --version
./f00-ls --version
```

*Figure. Build and gate flow*




**8. Procedure - quality gates**

8.1 Boring-solid (required before merge).

```
cd asm
make check
```

8.2 Hot path (real-work wall + CPU). Use when you change sort, ls, or other hot tools.

```
cd asm
make hot
```

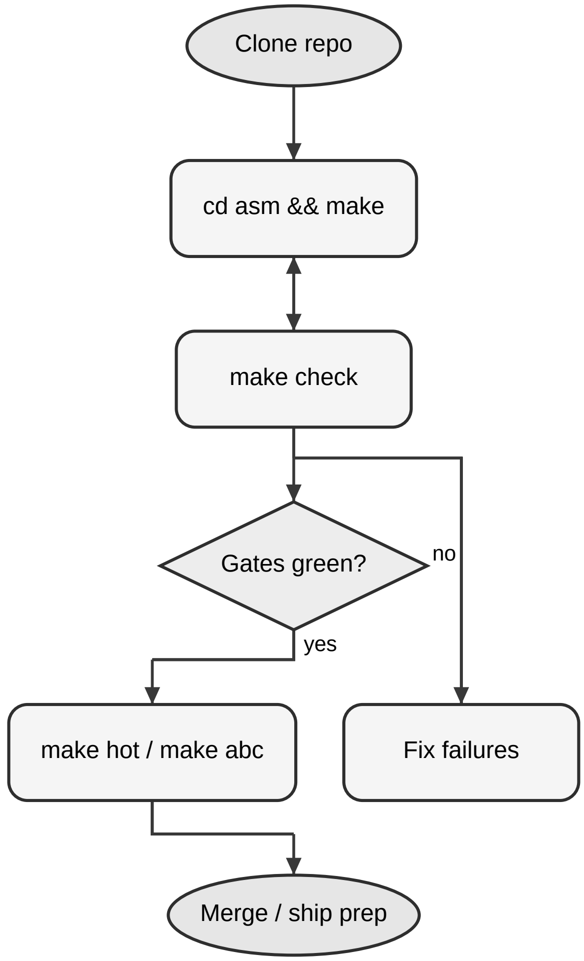
8.3 Speed law (wall + CPU). Optional each commit; required for law-2 claims. Includes hot.

```
cd asm
make speed
```

8.4 Structural speed proof (stress).

```
cd asm
make speed-stress
```

*Figure. Build and gate flow*




| **Target** | **Meaning** |
|---|---|
| make / make all | Build f00 + tool links |
| make smoke | Minimal runtime smoke |
| make parity | GNU --core parity battery |
| make check | smoke + parity (default bar) |
| make hot | Real-work wall+CPU + stdout assert |
| make speed | wall+CPU law (includes hot) |
| make speed-stress | Repeated full-speed-gate proof |
| make aarch64 | Build aarch64 multicall |
| make aarch64-smoke | qemu smoke of aarch64 binary |

### 9. Procedure - aarch64 port

The aarch64 path is shipped and grows toward full surface. It is not the default CI/release install path today.

9.1 Install the cross assembler and linker, and qemu-user if you build on x86-64.

9.2 Build and smoke.

```
cd asm
make aarch64
make aarch64-smoke
```

9.3 On failure, fix the port under asm/port/aarch64. Do not weaken x86-64 gates to paper over port gaps.

### 10. Procedure - local and end-user install

Default installer places multicall f00, all f00-* names, and bare tool names in INSTALL_DIR so replace wins on PATH.

10.1 Install from a local build.

```
cd asm
make install
# or
F00_LOCAL=/path/to/f00/asm bash /path/to/f00/install.sh
```

10.2 End-user install (primary ship story).

```
curl -fsSL https://f00.sh/install.sh | bash
```

10.3 Knobs: F00_VERSION=vX.Y.Z (pin tag); F00_SUPERSEDE=0 (f00-* only); INSTALL_DIR=~/.local/bin; F00_LOCAL=...; F00_MAN=1 (man pages, default on).

10.4 Config.

```
f00
f00-config
f00-config replace off
```

### 11. Procedure - keep installers identical

Root install.sh is the source of truth. Site and scripts copies must match byte-for-byte. Run after any edit to install.sh and before release.

11.1 Sync from repository root.

```
make sync-install
```

### 11A. Procedure - regenerate screenshots (every release)

Public screenshots must match the shipped binary and explain features (multi-command sessions), not decoration one-liners. Capture uses a real PTY and forced color environment.

11A.1 Build: cd asm && make (binary version string must match the release).

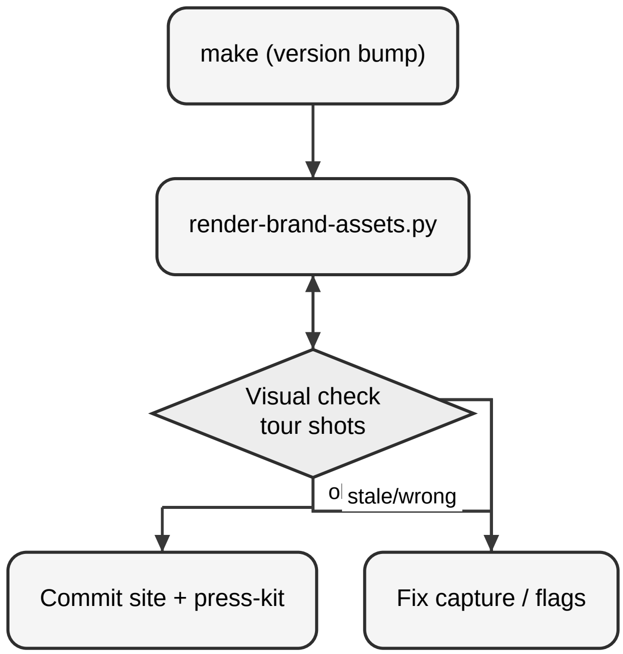
11A.2 Regenerate: python3 scripts/render-brand-assets.py

11A.3 Confirm outputs under site/assets/screenshots/, press-kit/screenshots/, docs/images/.

11A.4 Confirm site Feature tour (#tour) captions still match the shot set.

11A.5 Commit screenshots with the release; never leave stale version strings in hero.png.

*Figure. Screenshot regen on ship*




**12. Procedure - release and ship**

Primary release story: GitHub Release tarball + install.sh. deb/rpm/AUR/brew are secondary. Release assets must include file_id.diz, sop-f00tils-ops.pdf, and memo-release-VERSION.pdf (enforced by .github/workflows/release.yml).

12.1 Ensure main is green: build, make check, and speed gates as required for the change set.

12.2 Update version metadata so the binary, man pages, README pin examples, and CHANGELOG agree on vMAJOR.MINOR.PATCH.

12.3 Move Unreleased notes in CHANGELOG.md into a dated version section.

12.4 Refresh root file_id.diz for this version. Commit it. Attach it as a GitHub Release asset. Do not spotlight it on the website.

12.5 Sync installers (make sync-install).

12.6 Confirm man page(s), README.md, and site match behavior.

12.7 Create an annotated tag and push it.

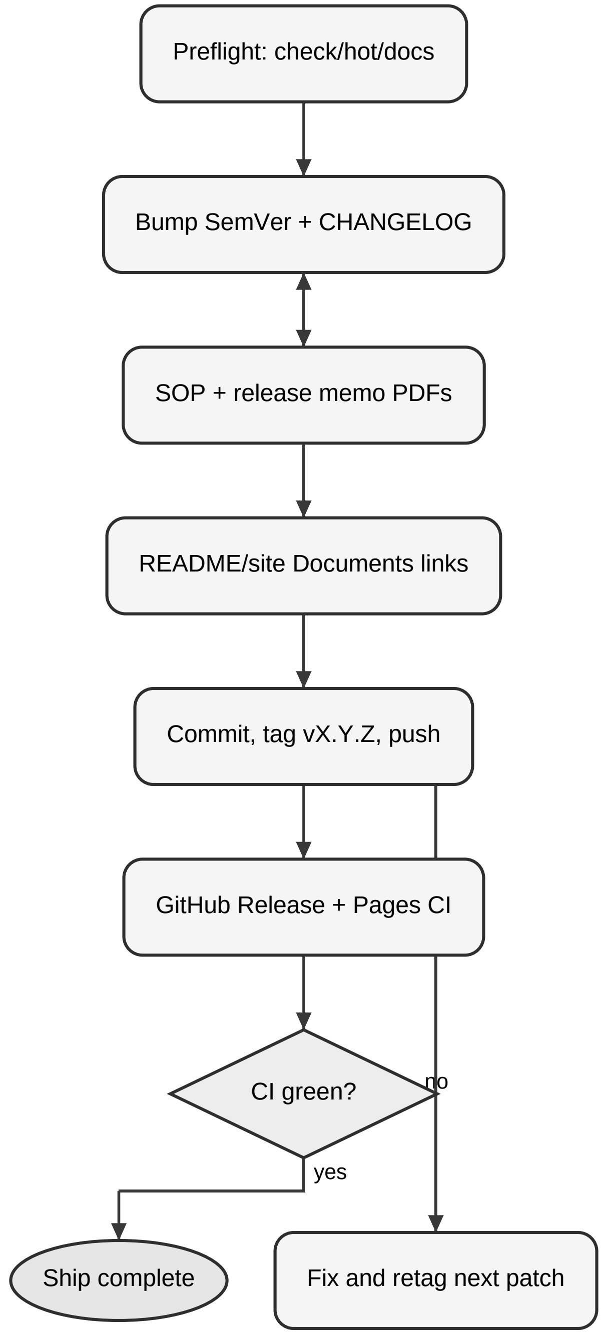
```
git tag -a vX.Y.Z -m "vX.Y.Z"
git push origin vX.Y.Z
```

12.8 Release workflow (tag v*) builds Linux x86-64 ASM, runs smoke/parity/speed-gate, verifies binary version, packages tarball, publishes GitHub Release.

12.9 Smoke-test the public install path. Do not move or reuse a published tag for different bits.

```
curl -fsSL https://f00.sh/install.sh | F00_VERSION=vX.Y.Z bash
f00 --version
```

*Figure. Release and deploy flow*




## 13. Procedure - secondary packaging

Use packaging scripts under scripts/ and configs under packaging/ and Formula/ when you intentionally ship secondary channels. They must not replace tarball + install.sh as the default install story.

- scripts/build-linux-packages.sh
- scripts/gen-aur-pkgbuild.sh / scripts/publish-aur.sh
- scripts/gen-homebrew-formula.sh / scripts/publish-homebrew-tap.sh
- scripts/sync-package-manifests.sh

### 14. Procedure - GitHub Actions constraints

- Use only Node 24 JS action majors (checkout@v6+, upload-artifact@v6+, download-artifact@v7+, softprops/action-gh-release@v3+, Pages @v5/@v6 as house rules specify).
- Do not reintroduce Node 20-era pins (checkout@v4, upload-artifact@v4, action-gh-release@v2). Deprecation annotations are a defect.

### 15. Procedure - user-facing text

- Name the project f00tils in narrative. Keep command names f00 / f00-*.
- Procedures, man pages, and install how-to: ASD-STE100 (Simplified Technical English).
- Public story (README lead, website): NASA Stylebook + AP habits.
- Ship narrative surfaces: README + https://f00.sh. Docs under docs/ are optional depth.

### 16. Daily operator checklist

- Branch from main; keep PRs small and Conventional Commits-friendly.
- Edit only assembly under asm/ for product logic.
- cd asm && make && make check before you open or update a PR.
- Run make hot / make speed when the change touches core-path performance or hot tools.
- Update man pages and site/README when user-visible behavior changes.
- Never claim --core parity without parity evidence.
- Never claim a speed win without gate evidence.

### 17. Failure responses

| Symptom | Action |
|---|---|
| make fails | Fix asm build. Confirm nasm/ld. Clean with make clean. |
| make check fails | Read smoke/parity logs. Restore --core GNU match. |
| make hot/speed fails | Profile the tool. Do not ship a slower core path. |
| install.sh mismatch | Run make sync-install. Re-diff site and scripts. |
| Release CI red | Match tag to binary version. Re-run gates locally. |
| aarch64 smoke fail | Fix port. Keep x86-64 gates independent. |
| Wrong action majors | Upgrade to Node 24-era pins per house rules. |

## 18. References

- AGENTS.md - product laws and layout
- README.md - public install and modes
- CONTRIBUTING.md - contributor build bar
- asm/Makefile - build and gate targets
- install.sh - primary installer
- .github/workflows/release.yml - tag release pipeline
- https://f00.sh - site, scoreboard, install entry
- site/bench/suite.json - suite bench data
- ~/.grok/skills/memo-pdf - house memo PDF renderer used for this document

## 19. Approval and control

This SOP is controlled under docs/. Update it when gates, install paths, or release mechanics change. Prefer short procedural edits over narrative expansion. Regenerate the PDF with the memo-pdf skill (nasa-sop type); do not freestyle layout.

Project Engineering
f00tils / OPS
Document: SOP-F00-OPS-001

Date: 2026-07-27

Enclosure: None. Companion sources are in-repository paths listed in section 18.
cc: Maintainers, Release Operators